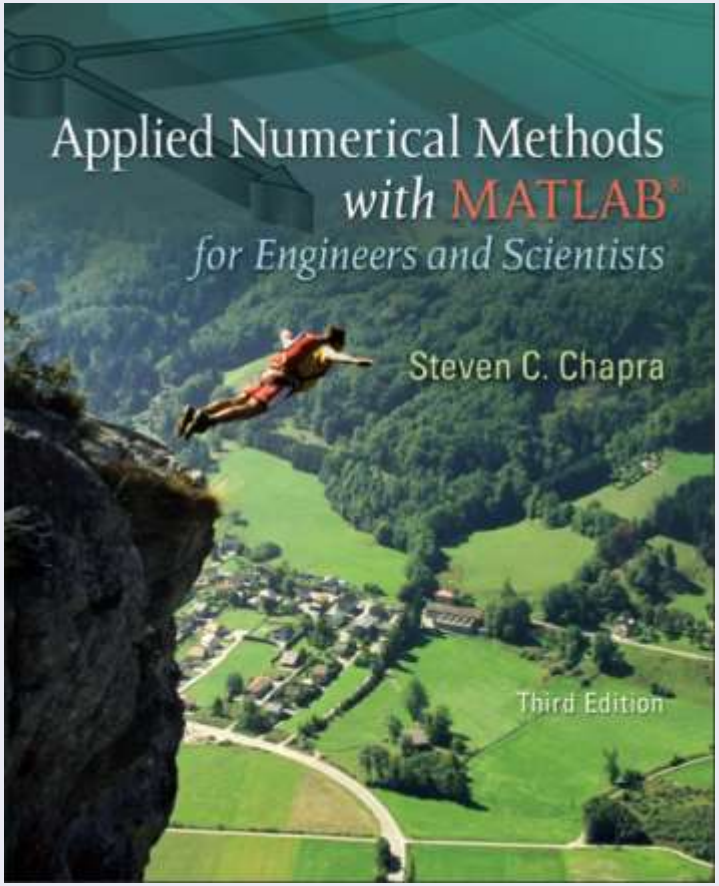


Part 1
Chapter 4

Roundoff and Truncation Errors

PowerPoints organized by Dr. Michael R. Gustafson II, Duke University and Prof. Steve Chapra, Tufts University

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Chapter Objectives

- Understanding the distinction between **accuracy** and **precision**.
- Learning how to **quantify error**.
- Learning how error estimates can be used to decide **when to terminate** an iterative calculation.
- Understanding how **roundoff errors** occur because digital **computers have a limited ability** to represent numbers.
- Understanding why floating-point numbers have limits on their range and precision.

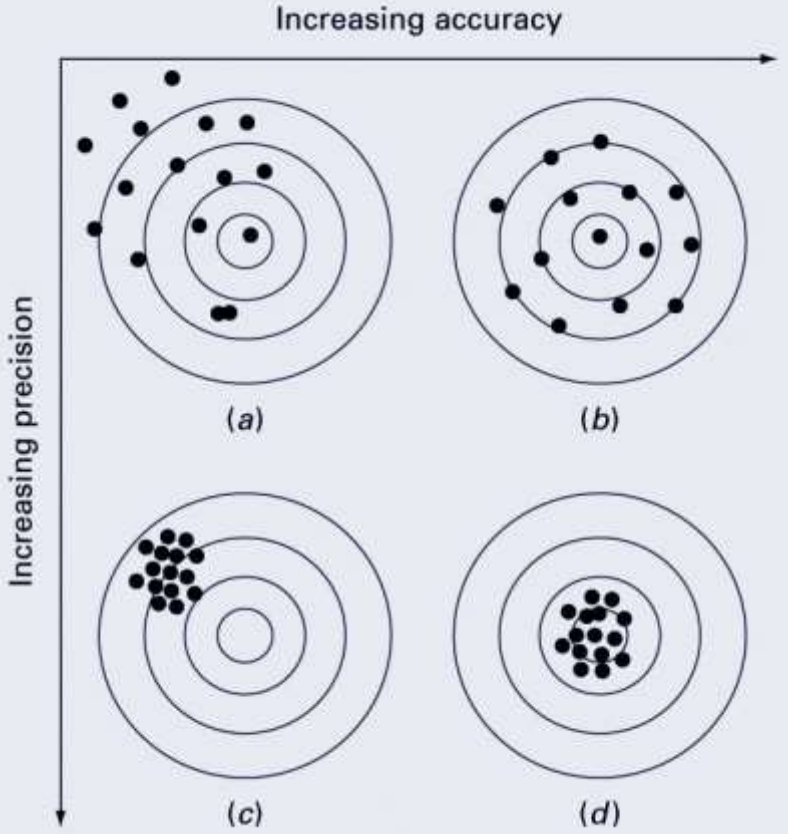
Objectives (cont)

- Recognizing that **truncation errors** occur when **exact mathematical formulations** are represented by approximations.
- Knowing how to **use the Taylor series to estimate truncation errors.**
- Understanding how to write forward, backward, and centered finite-difference **approximations of the first and second derivatives.**
- Recognizing that efforts to minimize truncation errors can sometimes increase roundoff errors.

Accuracy and Precision

- ***Accuracy*** refers to how closely a computed or measured value agrees with the true value, while *precision* refers to how **closely** individual computed or measured values **agree with each other**.

a) inaccurate and imprecise
b) accurate and imprecise
c) inaccurate and precise
d) accurate and precise



Error Definitions

- **True error (E_t):** the difference between the true value and the approximation.
- **Absolute error ($|E_t|$):** the absolute difference between the true value and the approximation.
- **True fractional relative error:** the true error divided by the true value.
- **Relative error (ε_t):** the true fractional relative error expressed **as a percentage.**

Error Definitions (cont)

- The previous definitions of error **relied on knowing a true value. If that is not the case, approximations** can be made to the error.
- **The approximate percent relative error** can be given as the **approximate error divided by the approximation**, expressed as a percentage - though this presents **the challenge of finding the approximate error!**
- **For iterative processes,** the error can be approximated as the difference in values between sucessive iterations.

Using Error Estimates

- Often, when performing calculations, we may **not be concerned with the sign of the error** but are interested in whether the absolute value of the percent relative error is lower than a prespecified **tolerance ε_s.** For such cases, the computation is repeated until

$$|\varepsilon_a| < \varepsilon_s$$

- This relationship is referred to as a ***stopping criterion*.**

Roundoff Errors

- *Roundoff errors* arise **because digital computers cannot represent some quantities exactly.** There are two major facets of roundoff errors involved in numerical calculations:
  - Digital computers have **size** and **precision limits** on their ability to represent numbers.
  - Certain numerical manipulations are highly sensitive to roundoff errors.

Computer Number Representation

- By default, MATLAB has adopted the **IEEE double-precision format** in which eight bytes (64 bits) are used to represent floating-point numbers:
  $n = \pm(1+f) \times 2^e$
- The sign is determined by a sign bit
- The mantissa *f* is determined by a 52-bit binary number
- The exponent *e* is determined by an 11-bit binary number, from which 1023 is subtracted to get *e*

Floating Point Ranges

- Values of –1023 and +1024 for *e* are reserved for special meanings, so the exponent range is –1022 to 1023.
- The largest possible number MATLAB can store has
  - *f* of all 1's, giving a significand of $2 - 2^{-52}$, or approximately 2
  - *e* of 1111111110_2, giving an exponent of 2046 – 1023 = 1023
  - This yields approximately $2^{1024} = 1.7997 \times 10^{308}$
- The smallest possible number MATLAB can store with full precision has
  - *f* of all 0's, giving a significand of 1
  - *e* of 0000000001_2, giving an exponent of 1–1023 = –1022
  - This yields $2^{-1022} = 2.2251 \times 10^{-308}$

Floating Point Precision

- The 52 bits for the mantissa *f* correspond to about 15 to 16 base-10 digits. The machine epsilon - the maximum relative error between a number and MATLAB's representation of that number, is thus
  $2^{-52} = 2.2204 \times 10^{-16}$

Roundoff Errors with Arithmetic Manipulations

- Roundoff error can happen in several circumstances other than just storing numbers - for example:
  - *Large computations* - if a process performs a large number of computations, roundoff errors may build up to become significant
  - *Adding a Large and a Small Number* - Since the small number's mantissa is shifted to the right to be the same scale as the large number, digits are lost
  - *Smearing* - Smearing occurs whenever the individual terms in a summation are larger than the summation itself.
    - $(x + 10^{-20}) - x = 10^{-20}$ mathematically, but
      x = 1; $(x + 10^{-20}) - x$ gives a 0 in MATLAB!

Truncation Errors

- *Truncation errors* are those that result from using an approximation in place of an exact mathematical procedure.
- Example 1: approximation to a derivative using a finite-difference equation:

$$\frac{dv}{dt} \cong \frac{\Delta v}{\Delta t} = \frac{v(t_{i+1}) - v(t_i)}{t_{i+1} - t_i}$$

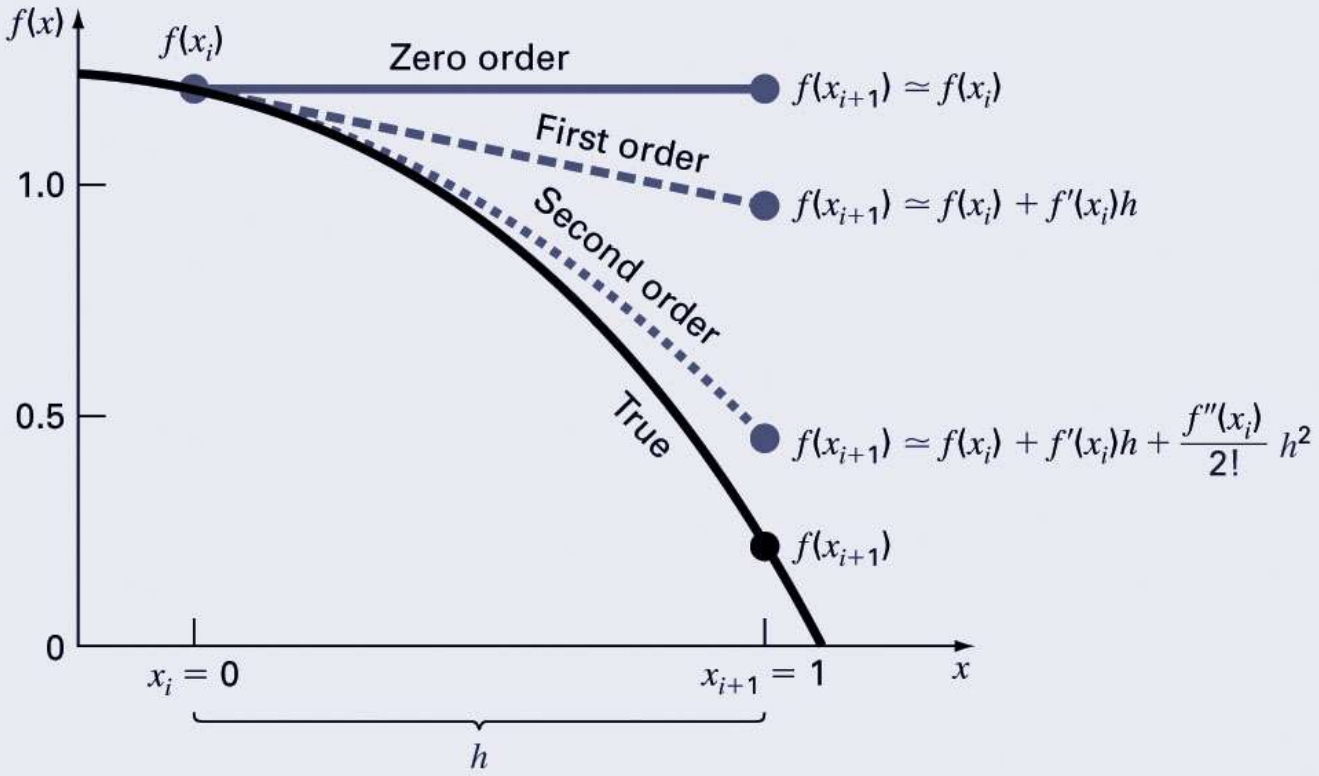
- Example 2: The Taylor Series

The Taylor Theorem and Series

- The *Taylor theorem* states that any smooth function can be approximated as a polynomial.
- The *Taylor series* provides a means to express this idea mathematically.

The Taylor Series

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \frac{f^{(3)}(x_i)}{3!}h^3 + \cdots + \frac{f^{(n)}(x_i)}{n!}h^n + R_n$$



Truncation Error

- In general, **the *n*th order Taylor series expansion will be exact for an *n*th order polynomial.**
- In other cases, the remainder term R_n is of the order of h^{n+1}, meaning:
  - The more terms are used, the smaller the error, and
  - The smaller the spacing, the smaller the error for a given number of terms.

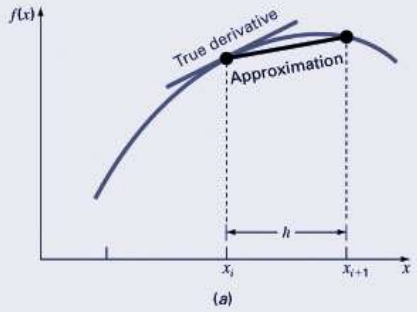
Numerical Differentiation

- The first order Taylor series can be used to calculate approximations to derivatives:
  - Given: $f(x_{i+1}) = f(x_i) + f'(x_i)h + O(h^2)$
  - Then: $f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} + O(h)$
- This is termed a “forward” difference because it utilizes data at *i* and *i*+1 to estimate the derivative.

Differentiation (cont)

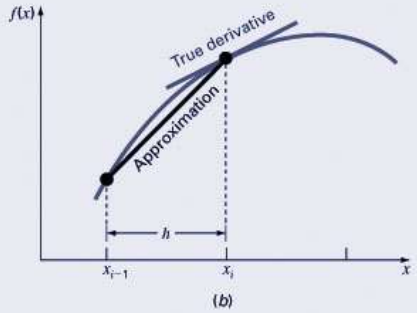
- There are also backward and centered difference approximations, depending on the points used:
- Forward:

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} + O(h)$$



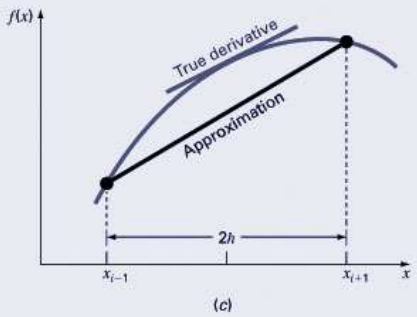
- Backward:

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h} + O(h)$$



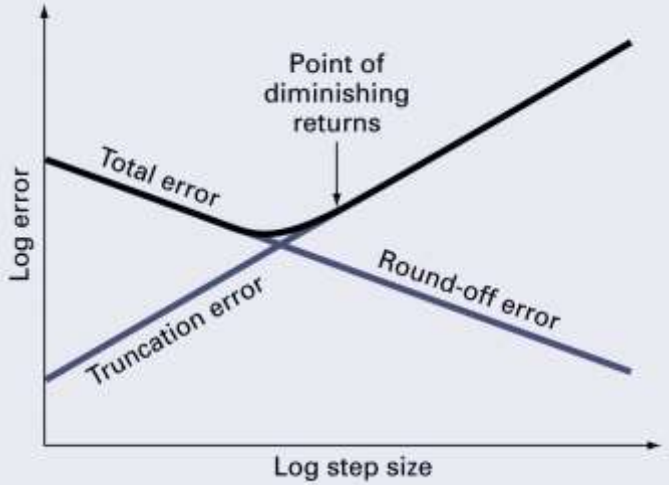
- Centered:

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h} + O(h^2)$$



Total Numerical Error

- The *total numerical error* is the summation of the truncation and roundoff errors.
- The truncation error generally *increases* as the step size increases, while the roundoff error *decreases* as the step size increases - this leads to a point of diminishing returns for step size.



Other Errors

- Blunders - errors caused by malfunctions of the computer or human imperfection.
- Model errors - errors resulting from incomplete mathematical models.
- Data uncertainty - errors resulting from the accuracy and/or precision of the data.